Project 5: Clinical trial matching AI Assistant

Concept

Help clinicians or patients find suitable clinical trials based on patient profiles, optimizing patient enrolment and research progress.

This system aims to bridge the gap between eligible patients and relevant clinical trials. It ingests comprehensive patient profiles from electronic health records (EHRs) and structured/unstructured data from clinical trial registries.

By leveraging a hybrid architecture of a Knowledge Graph and a Vector Database, the assistant can efficiently and accurately match patients based on precise structured criteria (e.g., diagnoses, demographics, lab values) and semantically interpret complex, nuanced textual information (e.g., symptoms described in clinical notes, detailed eligibility narratives in trial protocols).

The goal is to identify the most suitable trials, reduce screening failures, and accelerate patient recruitment for critical research.

Healthcare Value

- **Accelerated Research:** By streamlining patient recruitment, the system significantly reduces the time and cost associated with clinical trials, accelerating the development of new treatments.
- **Improved Patient Outcomes:** Connects patients to cutting-edge therapies and personalized treatment options that might not be available through standard care.
- **Reduced Burden on Clinicians:** Automates a complex and time-consuming manual process for clinicians, allowing them to focus more on patient care.
- **Increased Access to Trials:** Broadens the reach of clinical trials, making it easier for eligible patients, especially those with rare diseases or in underserved areas, to find and enroll in studies.drugs and diseases.

Key Features for Demo

1. Precise Structured Criteria Matching (Knowledge Graph):

- **Capability:** Rapidly identify trials matching exact patient attributes and explicit medical history.

- **Demonstration:** "Find all Phase III trials for patients aged 18-65, diagnosed with 'Type 2 Diabetes' (ICD-10 E11.9) within the last 5 years, who are currently on 'Metformin' and have a HbA1c between 7.0% and 9.0%."
- **Suggested Approach:** The KG excels at filtering based on exact values, ranges, and multi-hop relationships (Patient -> Diagnosis -> Medication -> Lab Result).

2. Semantic Eligibility Interpretation & Symptom Matching (Vector Database/RAG):

- **Capability:** Understand and match nuanced patient symptoms and disease characteristics described in free-text clinical notes against similarly nuanced eligibility narratives in trial protocols.
- **Demonstration:** "Identify trials for patients whose clinical notes describe 'persistent, debilitating fatigue' and 'unexplained muscle weakness' for at least 6 months, even if these specific phrases aren't codified as a formal diagnosis."
- **Sugegsted Approach:** The VectorDB (as part of a RAG system) semantically matches the patient's unstructured narrative with trial criteria text.

3. Complex Rule Evaluation & Contextual Matching (Hybrid - KG + Vector Database):

- **Capability:** Evaluate sophisticated inclusion/exclusion criteria that require combining structured facts with semantic understanding of unstructured text.
- **Demonstration:** "Match patients with 'advanced metastatic prostate cancer'
- **Suggested approach:** Find who all have *documented evidence of rapid disease progression in their recent oncology notes* (VectorDB/RAG) and who have *not* received prior treatment with any 'androgen receptor signaling inhibitors' (KG traversal)." This combines precise structured filtering with semantic evidence extraction.

4. Trial Suggestion & Prioritization (Hybrid - KG + Vector Database/ML):

- **Capability:** Move beyond strict matching to suggest trials that are a good *semantic fit* for a patient's overall profile or research interest, and prioritize them.
- **Demonstration:** "For a patient with a rare genetic disorder, suggest active trials that focus on similar genetic pathways or therapeutic targets, even if the trial is for a different primary diagnosis. Also, prioritize trials based on novelty of intervention, geographical proximity, and phase."
- **Suggested approach:** Leverage KG for pathway relationships and VectorDB for semantic similarity across research domains.

5. Dynamic Criteria and Knowledge Evolution (RAG Integration):

- **Capability:** Provide context and interpretation of trial criteria and medical terms by dynamically retrieving information from full protocol documents or external knowledge bases.
- **Demonstration:** "When a patient's lab result shows 'uncontrolled hypertension,' explain what 'uncontrolled' means specifically for *this particular trial's eligibility criteria* by retrieving relevant definitions or thresholds from the trial protocol document."
- **Suggested approach:** The RAG component allows the system to pull in and present precise textual context.

Data Sources

Generate synthetic datasets for above

Tip: Copy paste the above requirements and ask Claude or any other chatbot to create python script that generates the synthetic data. Also ask it to generate documents that simulate clinical trial publications. And ask it to validate the patterns as well

A reference synthetic data is available in the project5-data folder. It contains:

- patients_structured.json/csv: 100 patient records

- clinical_notes.json: 132 clinical notes

- trials_structured.json: 50 trial records

- trial_protocols.json: 6 protocol documents

- protocol_documents/: 6 individual protocol text files

Key Features in the dataset:

1. Structured Patient Data

- Demographics, diagnoses with ICD codes, medications, lab results
- Family history, smoking status, treatment history
- Realistic medical values (HbA1c for diabetes patients, etc.)

2. Unstructured Clinical Notes

- Progress notes with symptoms and physician observations
- Discharge summaries with disease progression details
- Realistic medical narratives that include phrases like "persistent fatigue," "rapid disease progression," etc.

3. Structured Trial Registry Data

- Trial phases, conditions, sponsors, locations
- Inclusion/exclusion criteria with age ranges, gender requirements
- Enrollment targets, study status, intervention types

4. Unstructured Trial Protocol Documents

- 6 detailed protocol documents with sections for:
    - Study objectives and design
    - Detailed inclusion/exclusion criteria
    - Investigational product descriptions
    - Safety considerations and adverse event reporting

Data Relationships for Testing:

The script creates realistic scenarios for testing your hybrid KG + Vector DB system:

- **Structured matching**: Patients with specific ICD codes, lab values, medications that match trial criteria
- **Semantic matching**: Clinical notes with symptoms like "debilitating fatigue" or "rapid disease progression" that need to be matched semantically
- **Complex rule evaluation**: Trials requiring both structured criteria (e.g., diabetes + HbA1c range) AND semantic evidence (e.g., "uncontrolled" condition described in notes)